

Foundations of AI 1: From Natural Language Processing to Large Language Models

by

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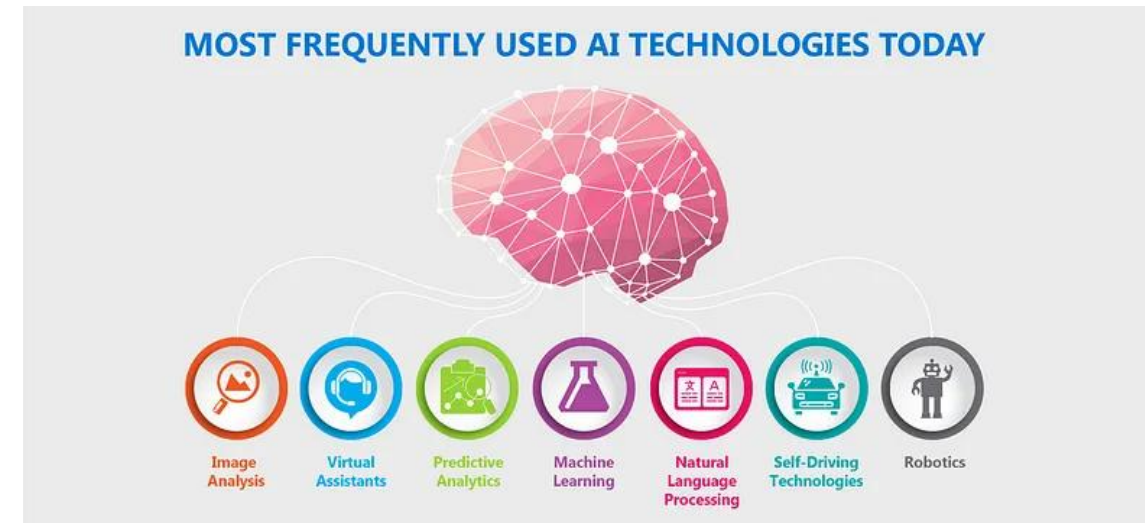
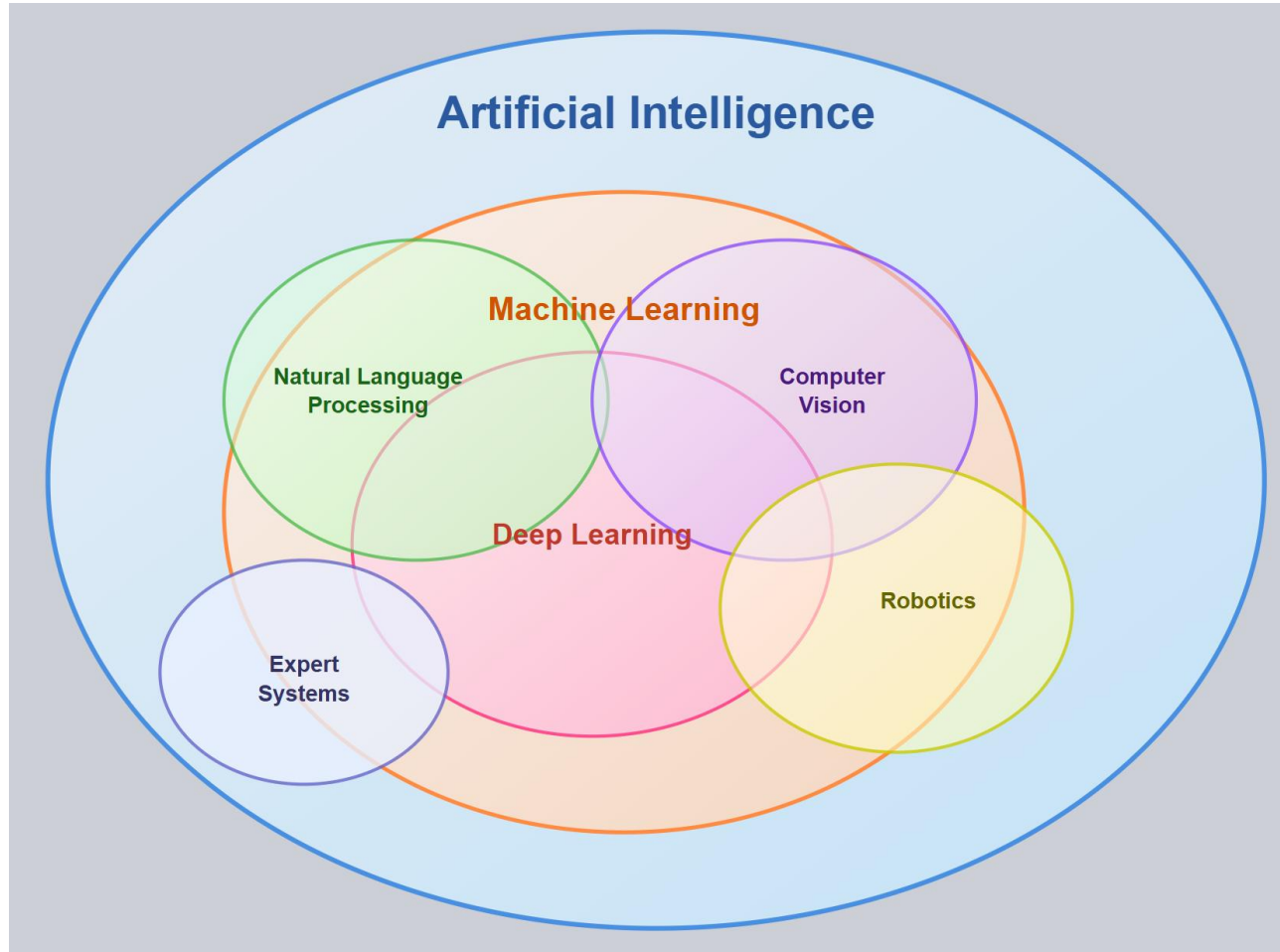
Dartmouth Libraries

Dartmouth

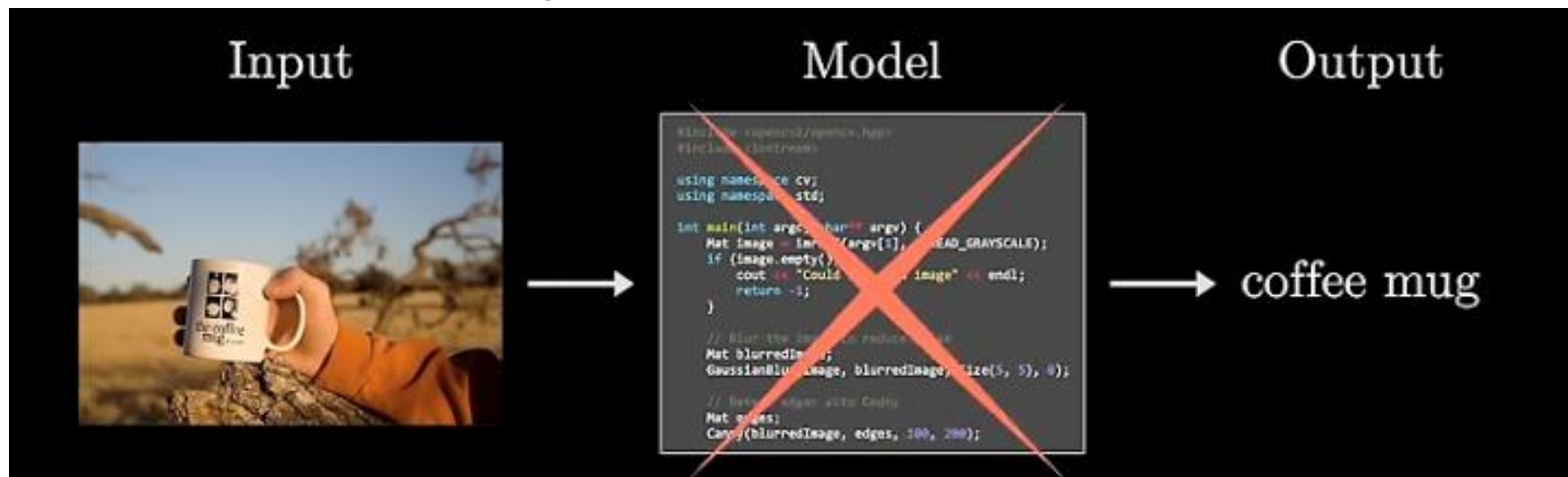
Mashrekur.Rahman@dartmouth.edu

1. Go to *jhub.dartmouth.edu*
2. Click *Start My Server*
3. Select *Reproducible Research Workshops*
4. *RR-workshops/foundations-of-ai*
5. Select *Notebook 0*
6. Click on *Kernel* and select “*Restart Kernel and Run All Cells*”

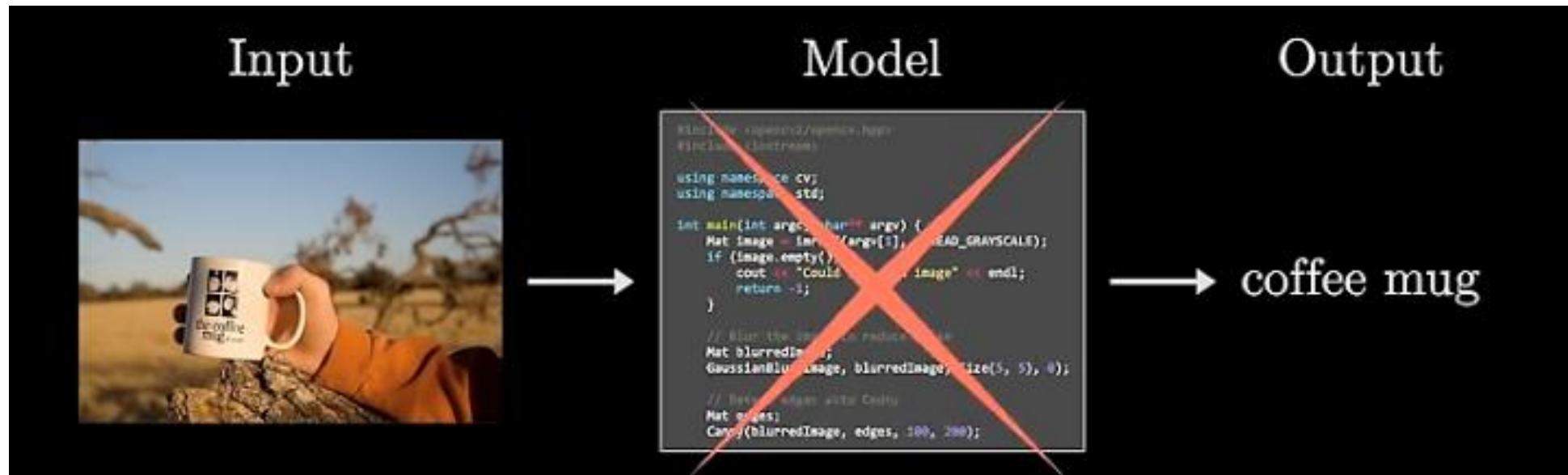
Let's dive into the basics of Artificial Intelligence...



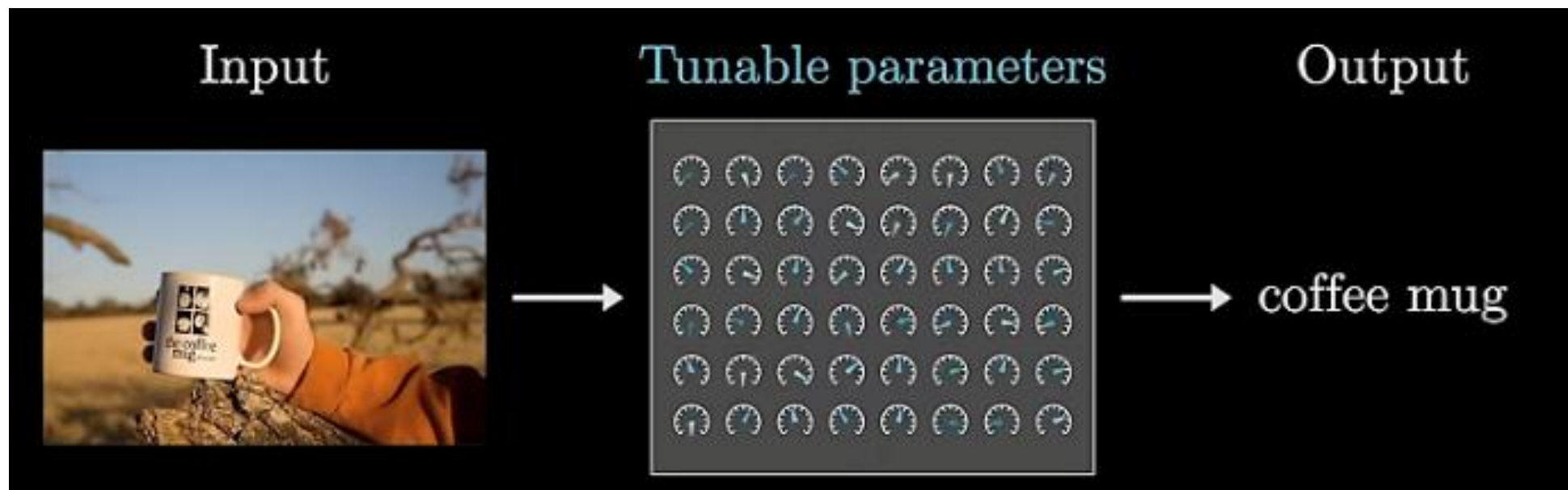
Traditional Modeling : Explicitly define how to do a task in code



Traditional Modeling : Explicitly define how to do a task in code



AI Modeling: Flexible structure that learns from data



Exercise 1:

AI models learn from data by minimizing errors between its predictions versus the reality!

Open Notebook 1: You are inside an AI model now – can you adjust the internal knobs and dials to minimize the model error?

Two Broad Types of AI Models: Count-Based vs. Network-Based – When to use?

Count-Based Models (Probabilistic)

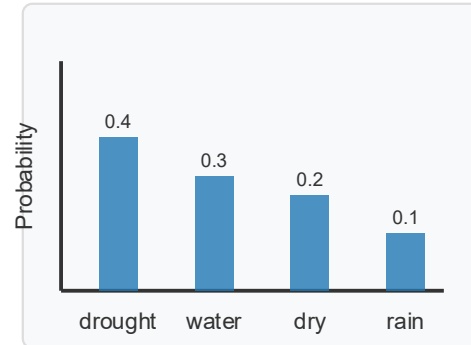
When to Use Count-Based Models

Choose When You Need:

- Transparency & Explainability
- Limited Data Available
- Structured Information
- Low Computational Resources

Research Use Cases:

- **Text Analysis**
Sentiment analysis, surveys
- **Medical Diagnostics**
Symptom-based predictions
- **Policy Research**
Document term analysis
- **Field Research**
Resource-limited settings



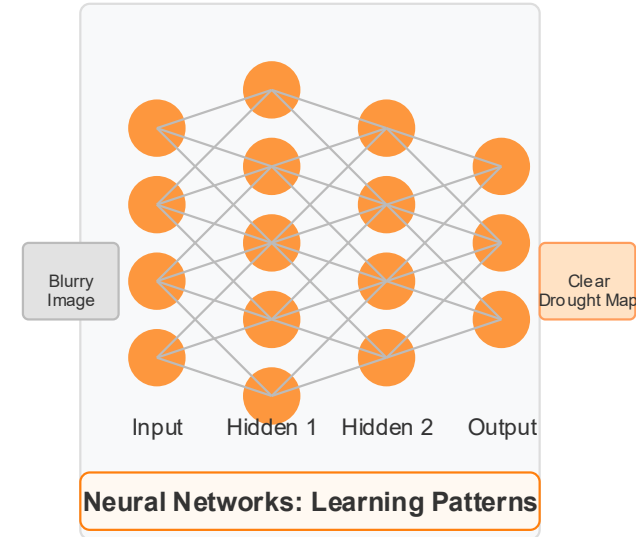
Bayes' Theorem at Work

40% chance this tweet signals concern

Examples

- Naive Bayes Classifiers
- Logistic Regression
- Term Frequency-Inverse Document Frequency (TF-IDF)
- Hidden Markov Models

Network-Based Models



Neural Networks: Learning Patterns

Examples

- Convolutional Neural Networks
- Recurrent Neural Networks
- Transformers
- U-Net Architectures

When to Use Network-Based Models

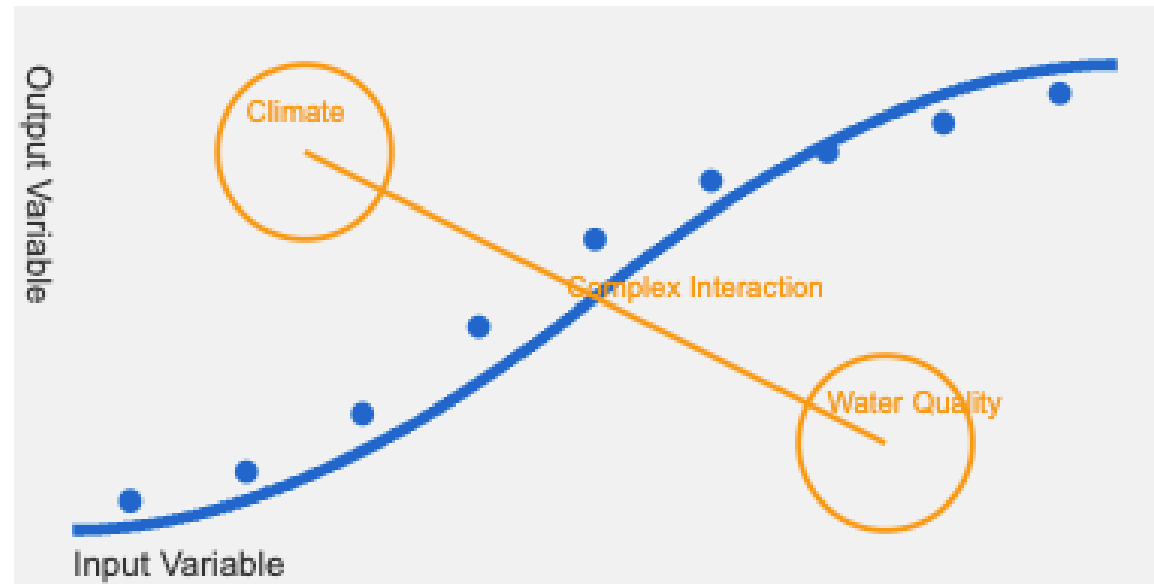
Choose When You Need:

- Complex Pattern Recognition
- Large Datasets Available
- Unstructured Data (images, etc.)
- High Computational Resources

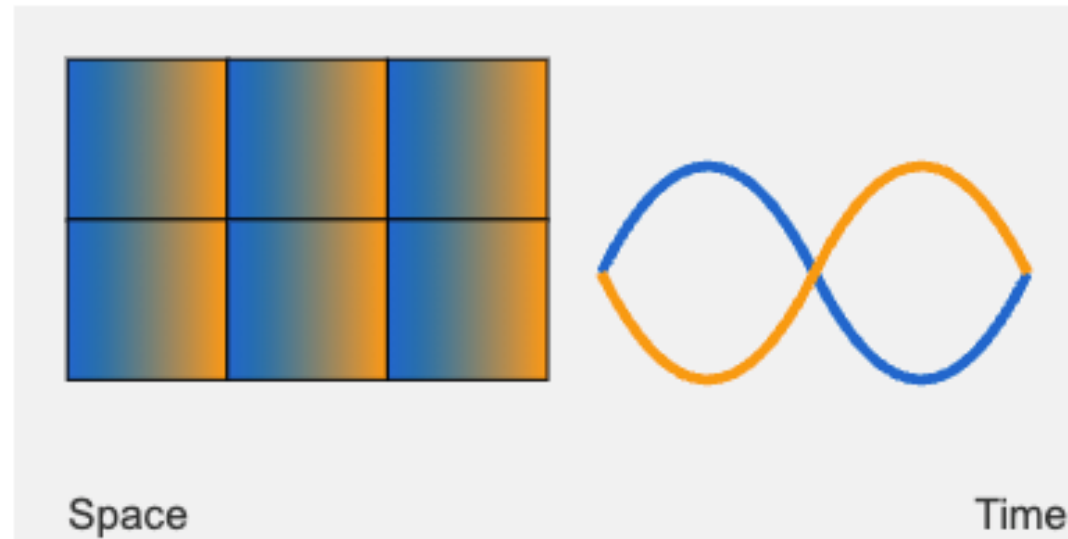
Research Use Cases:

- **Environmental Monitoring**
Satellite imagery analysis
- **Medical Imaging**
MRI, CT scan interpretation
- **Climate Modeling**
Time-series predictions
- **Cross-Disciplinary Text**
Complex literature analysis

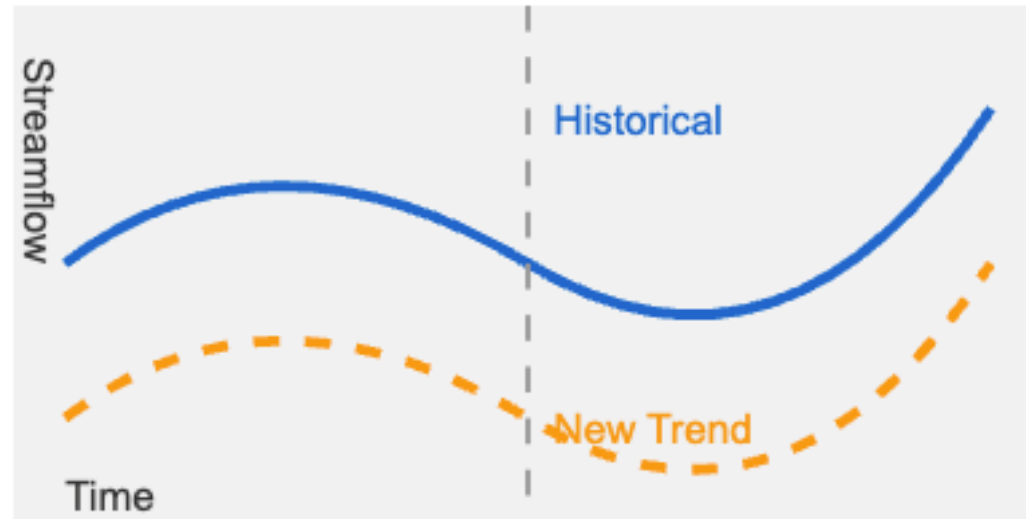
*We can capture the **Complex & Nonlinear** Relationships in Systems....*



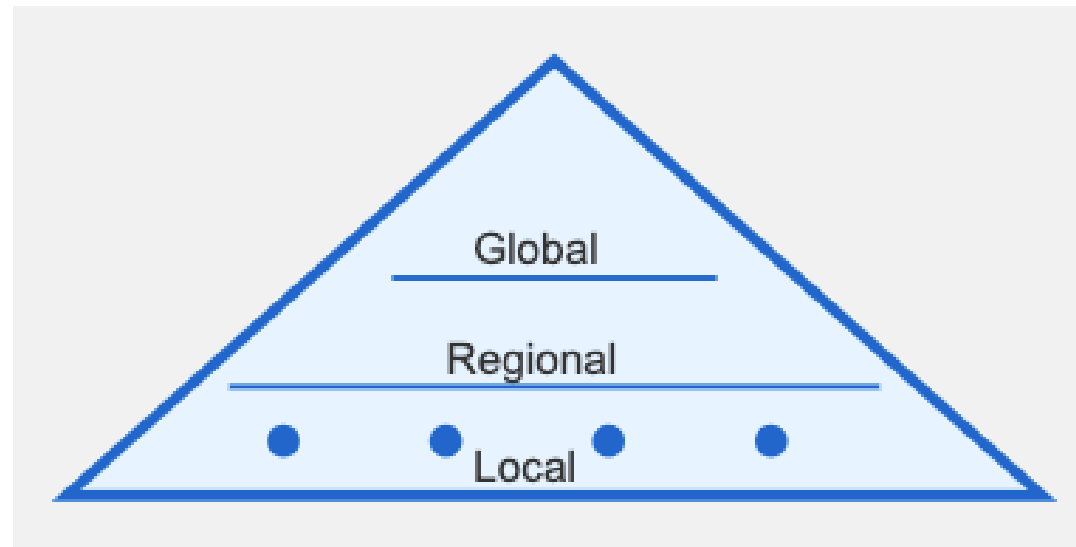
*We can learn from **Heterogeneity**...*



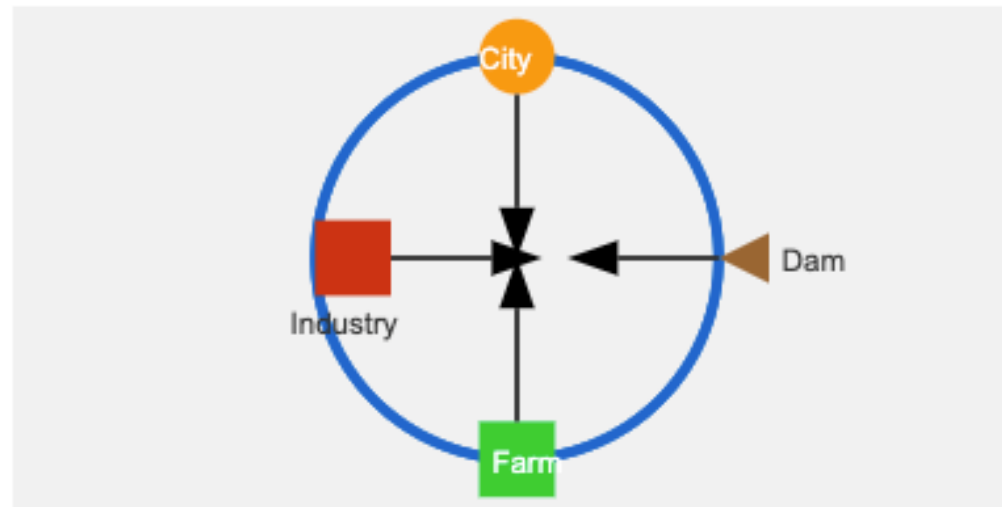
*AI systems can capture the **Nonstationarity**
in data....*



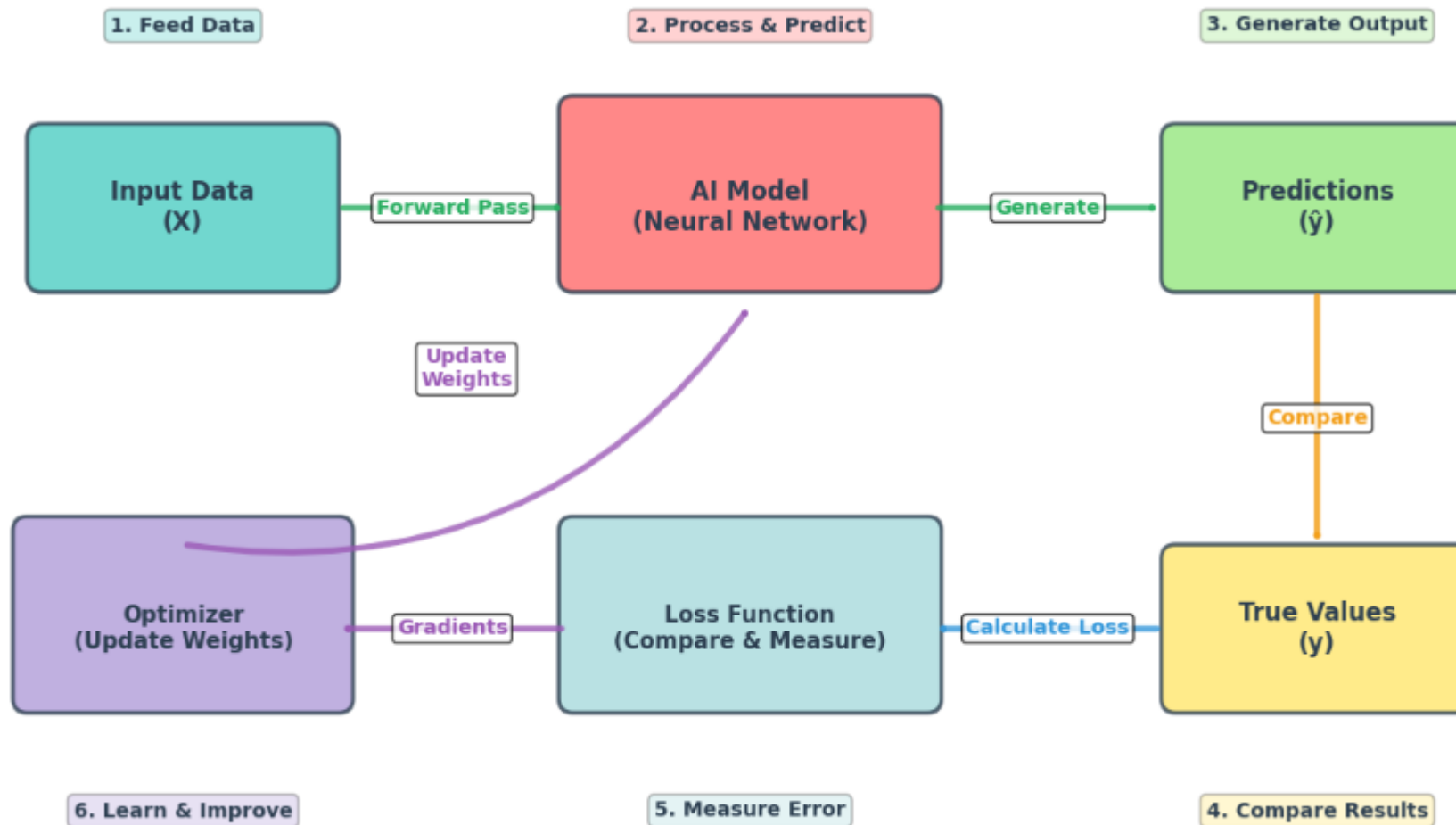
*AI Systems are **Scalable & Computationally Efficient**....*



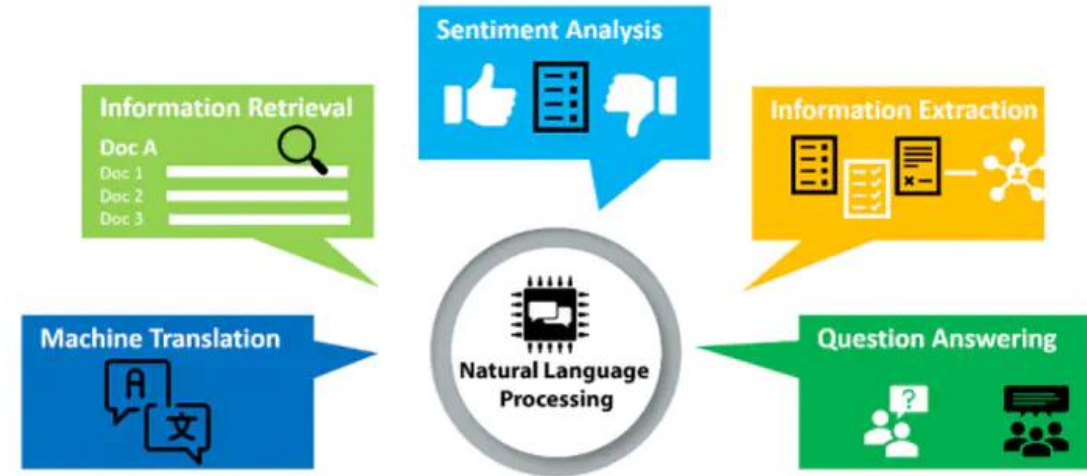
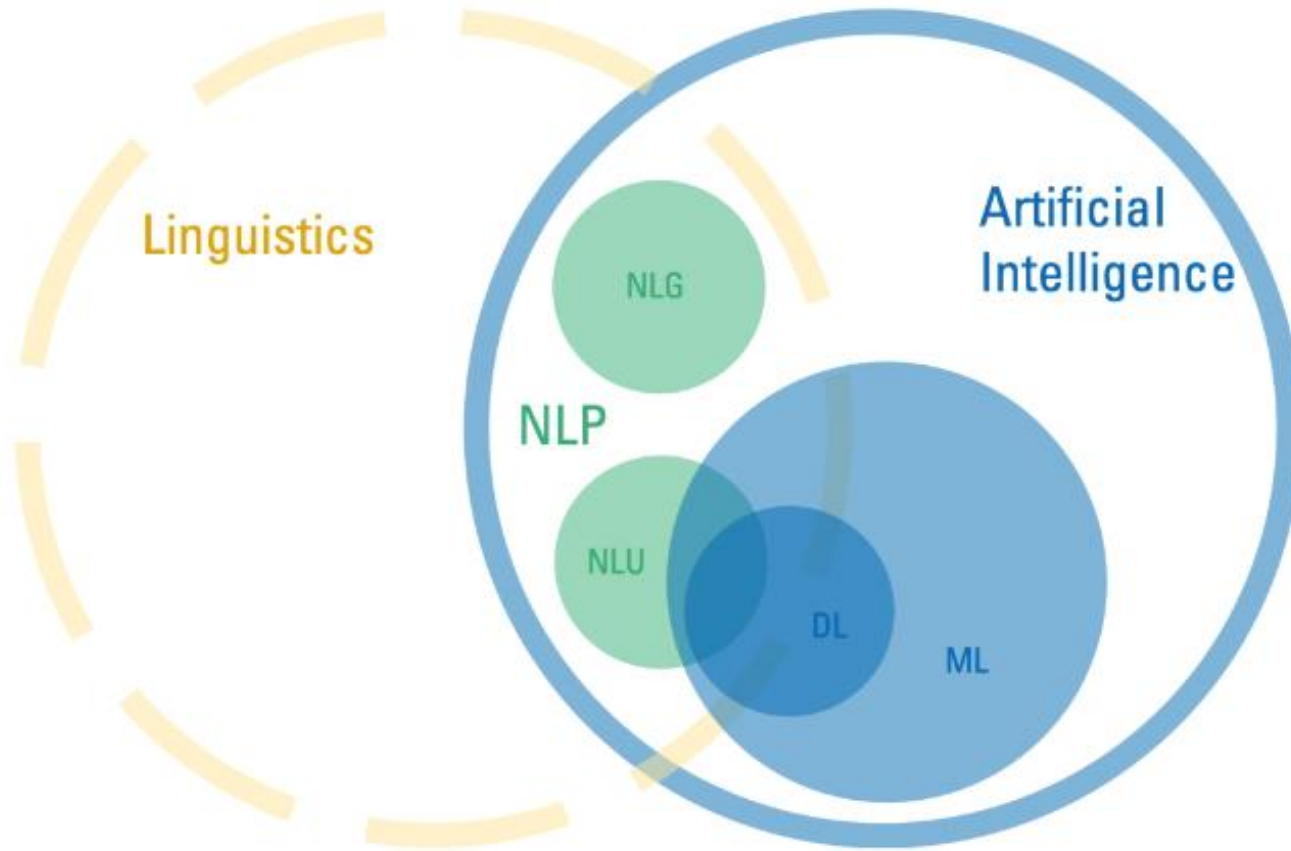
We can leverage many modes of data
Human-Environment Interactions...



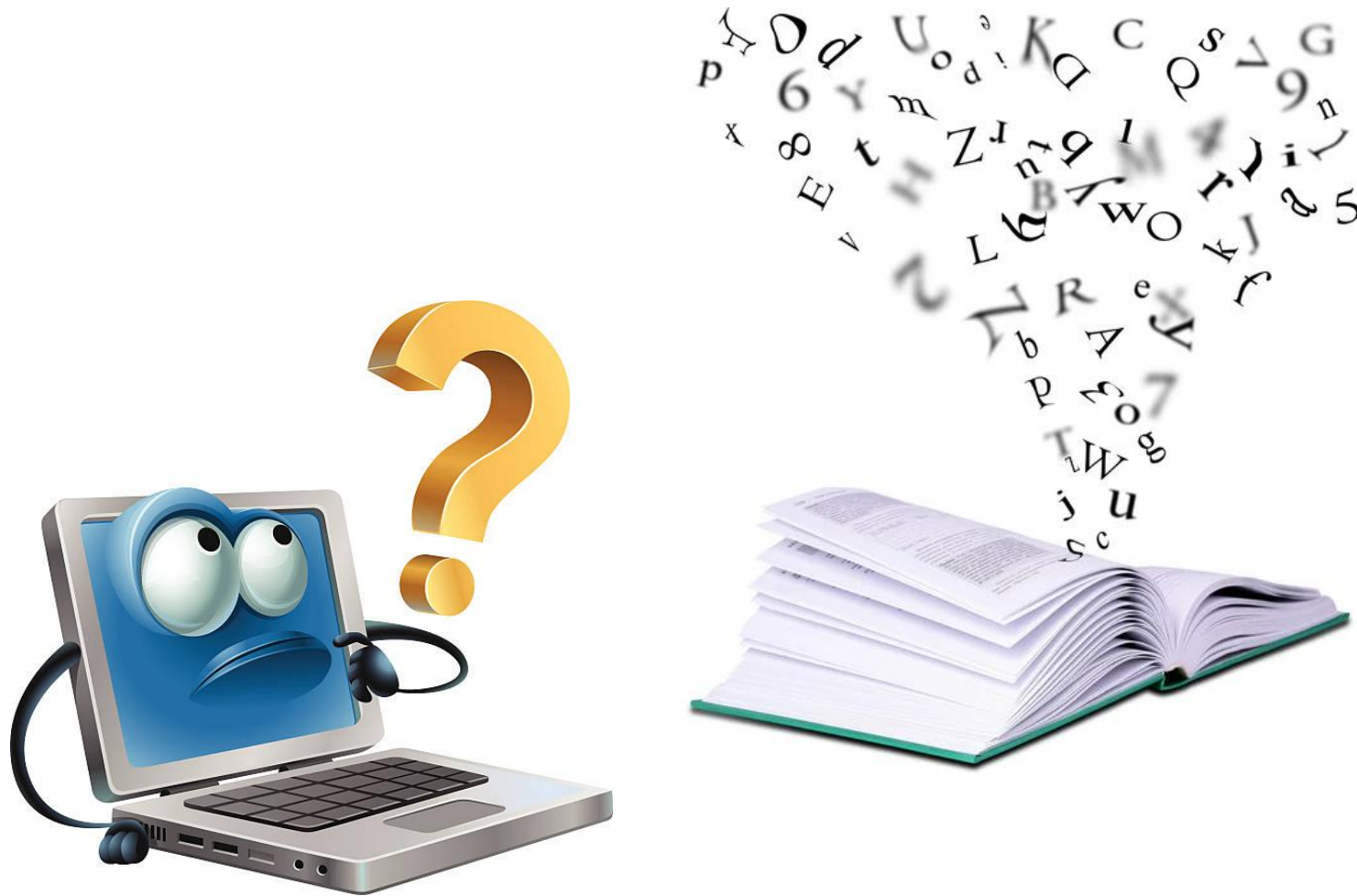
Exercise 2 (Notebook 2): Visualize the basic AI learning loop



Natural Language Processing – An Introduction



Foundations of NLP – How do computers understand human language?



Exercise 3:

Run all the kernels of notebook 3; visualize the differences between vector encodings & embeddings

The Fundamental Challenge of NLP

What Humans See

cat

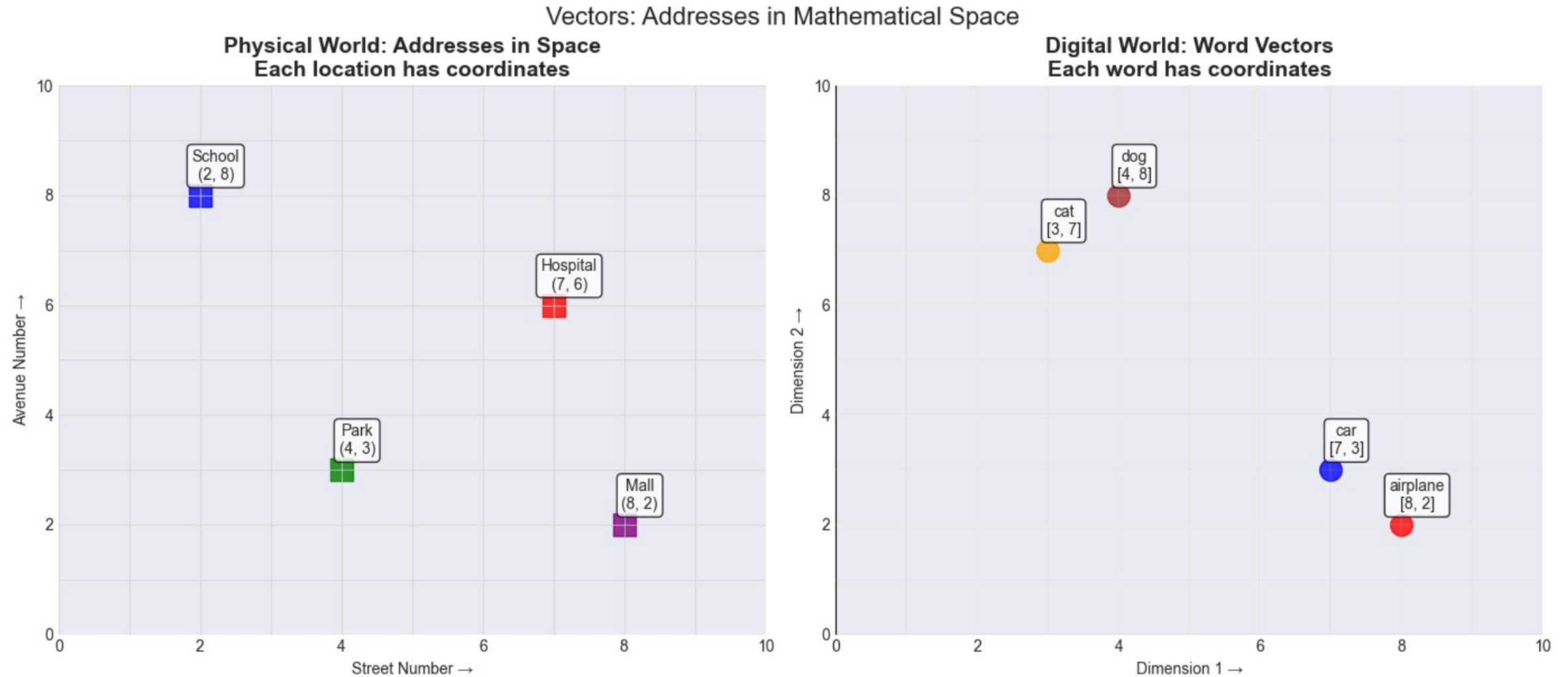
Furry friend

What Computers See

???

01100011 01100001 01110100

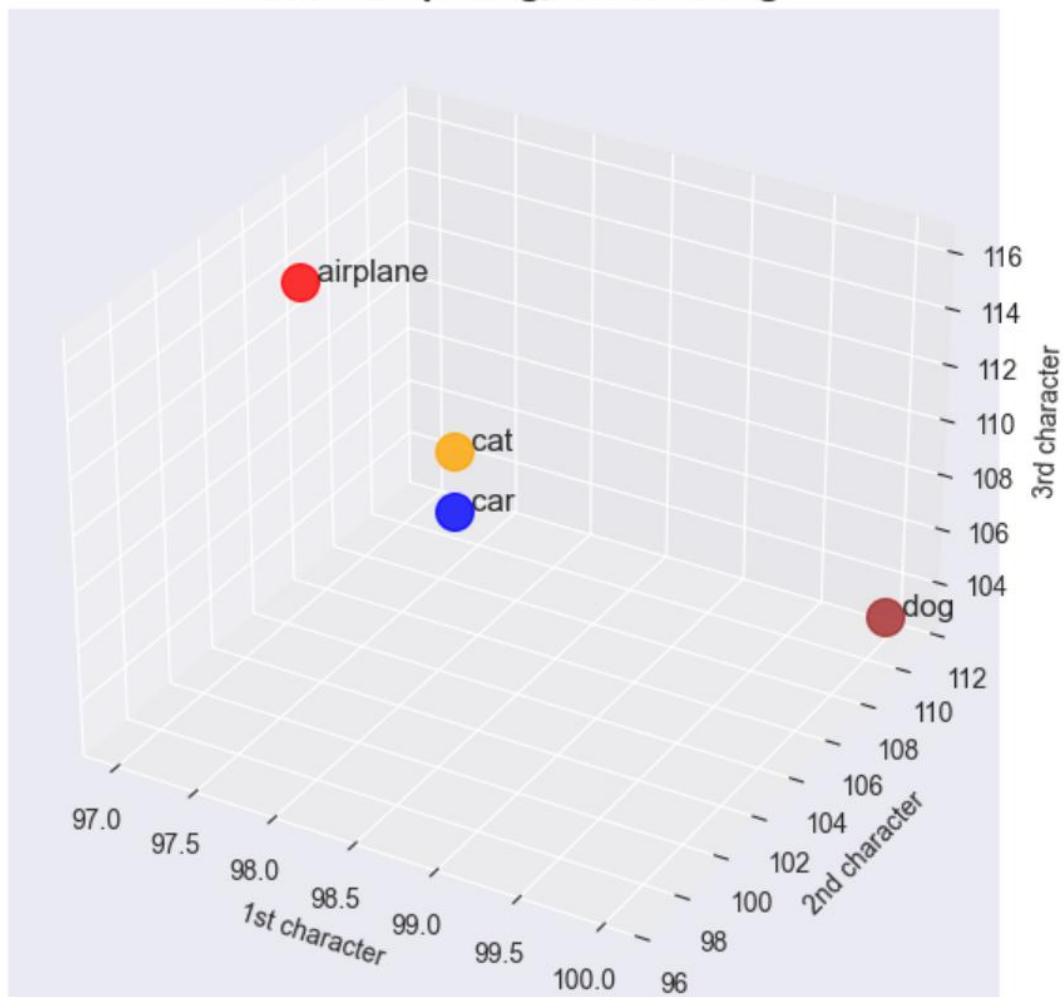
Vectors - represent human language as numbers so that computers can understand them



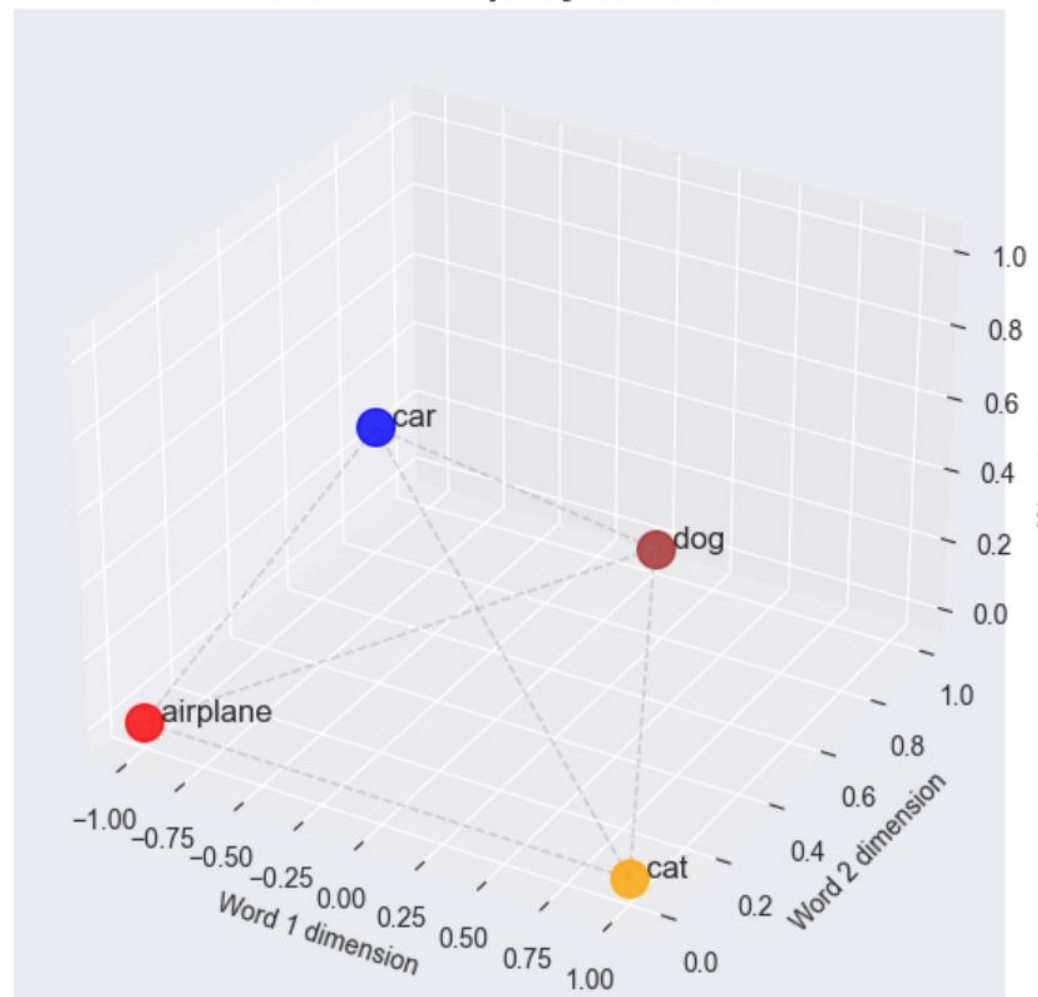
Traditional Encodings

Traditional Encodings: Numbers Without Meaning

ASCII ENCODING
Based on spelling, not meaning

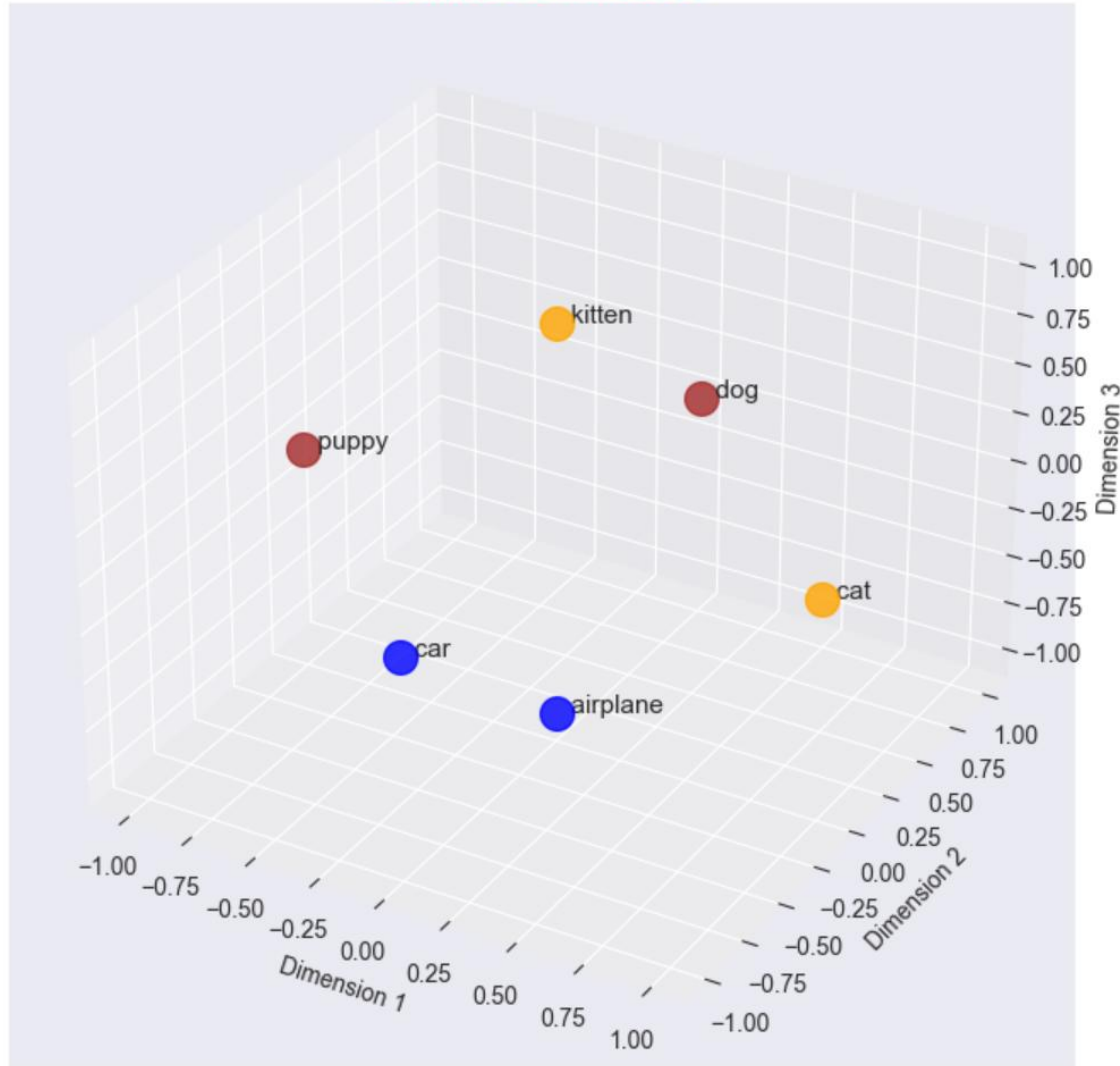


ONE-HOT ENCODING
All words equally different

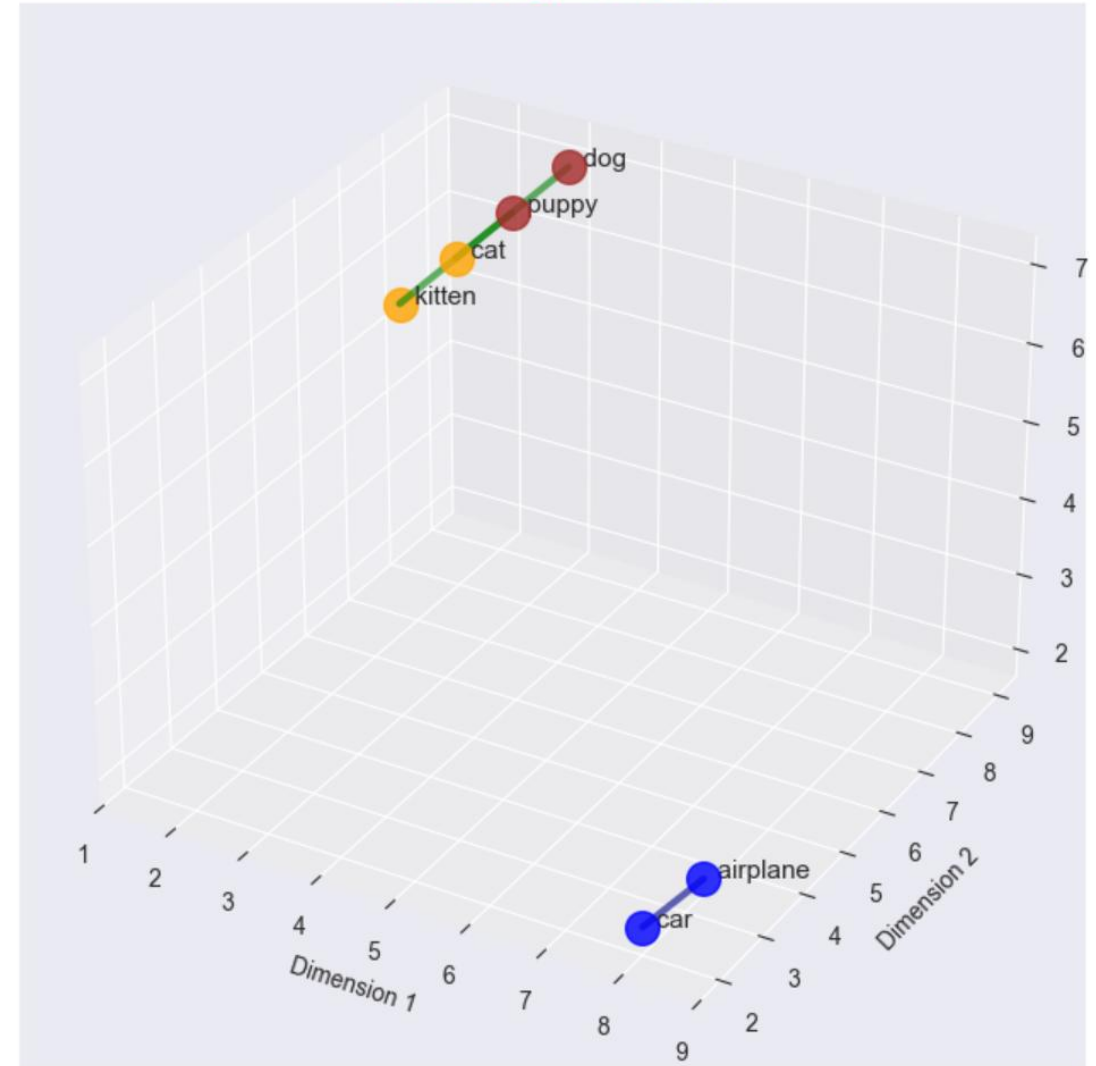


Vector Embeddings – Gives Words Meaning

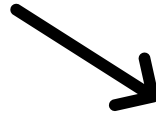
**Traditional Encodings:
Scattered Addresses**



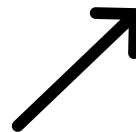
**Embeddings:
Smart Neighborhoods**



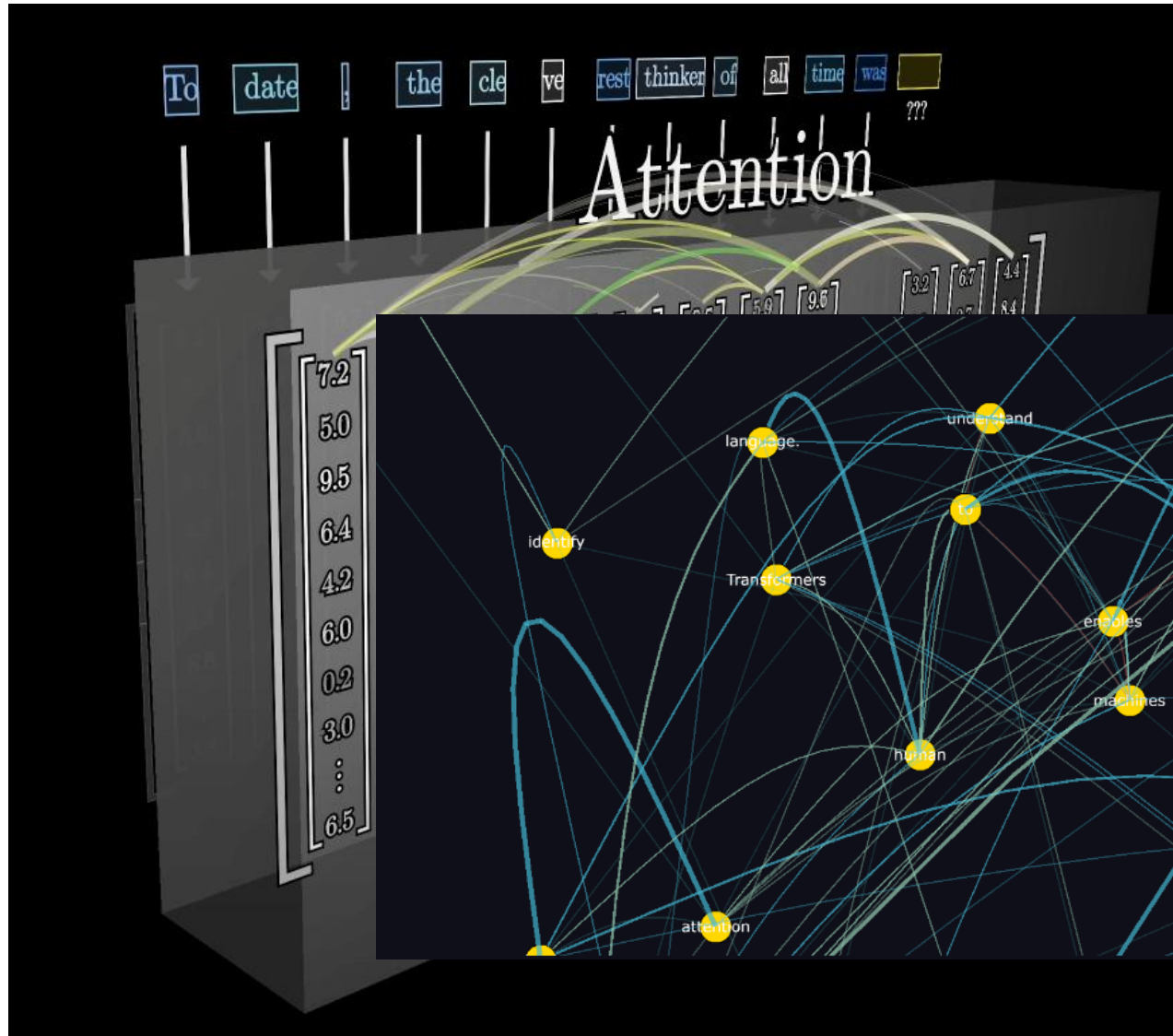
Vector Embeddings – Gamechanger



**Vector
Databases**



Attention is all you need....



Attention mechanism allows each element of the input data to **attend to** every other element of the input data.

Exercise 4 (Notebook 4):

Run notebook 4 – play with the intuitive visualization mimicking an Attention network

- What are the fundamental advantages of such a network?*
- Think about the task of predicting the next word, what does the Attention mechanism allow us to do?*

Diving deeper into Transformers...(we'll skip this for now)

Attention Is All You Need

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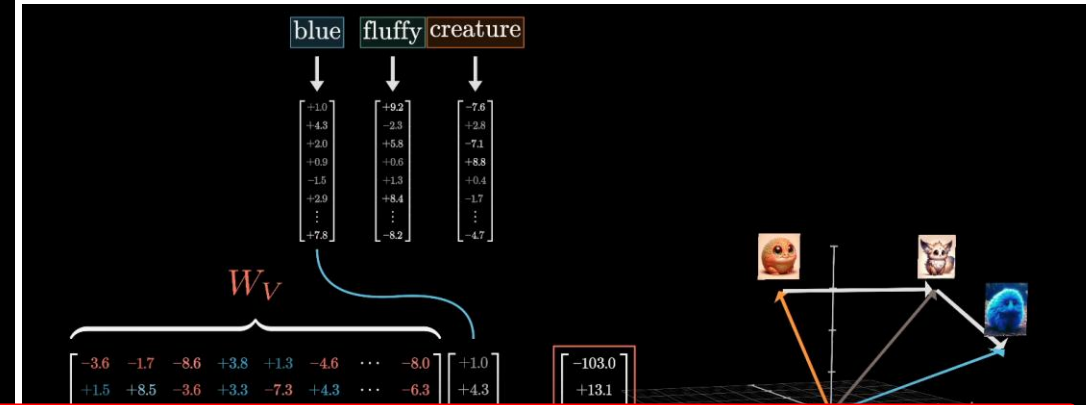
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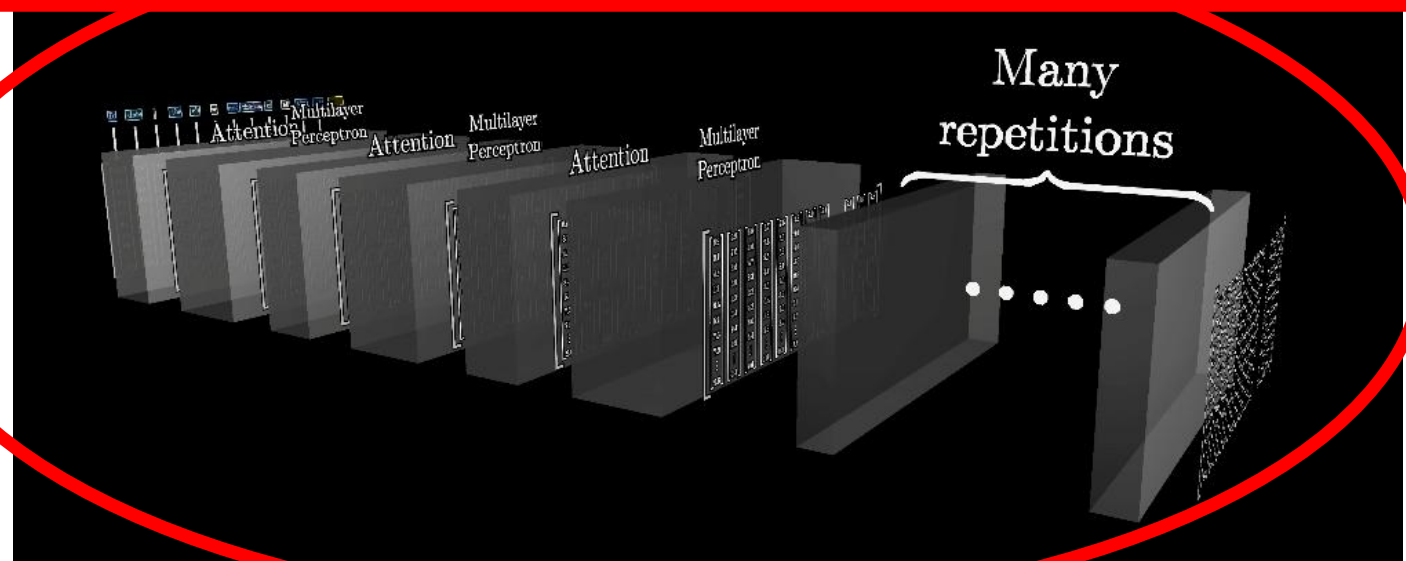
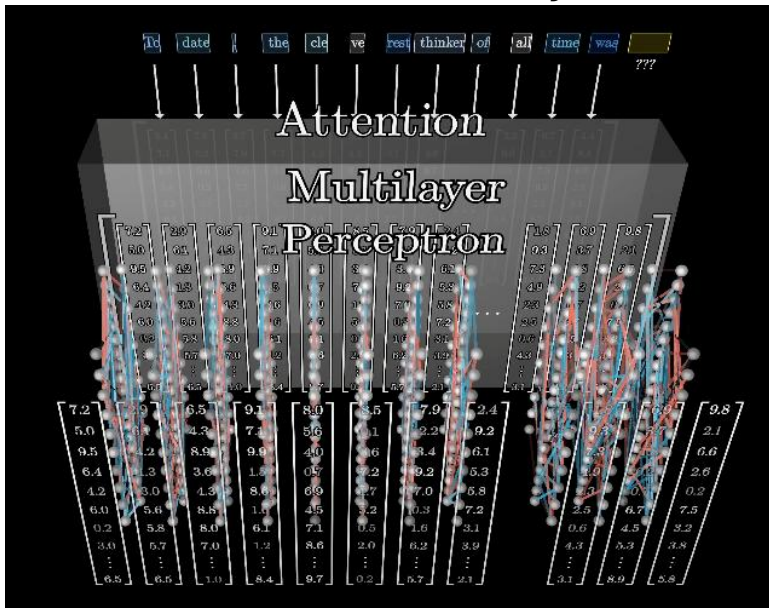
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

	Q_1	Q_2	Q_3	Q_4	Q_5	$\dots$	Q_n	
K_1	$\frac{Q_1 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_1}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_1}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_1}{\sqrt{d_k}}$	
K_2	$\frac{Q_1 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_2}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_2}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_2}{\sqrt{d_k}}$	
K_3	$\frac{Q_1 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_3}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_3}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_3}{\sqrt{d_k}}$	
K_4	$\frac{Q_1 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_4}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_4}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_4}{\sqrt{d_k}}$	
K_5	$\frac{Q_1 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_2 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_3 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_4 \cdot K_5}{\sqrt{d_k}}$	$\frac{Q_5 \cdot K_5}{\sqrt{d_k}}$	$\dots$	$\frac{Q_n \cdot K_5}{\sqrt{d_k}}$	



- Can capture nonlinear dependencies
- Each transformer block has multiple attention heads
- There are multiple transformer blocks in a model

Each attention head is followed by an MLP...

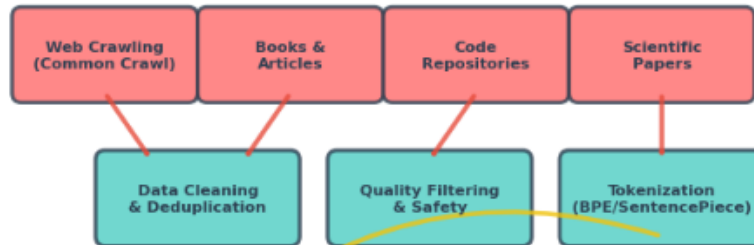


Courtesy: 3Blue1Brown

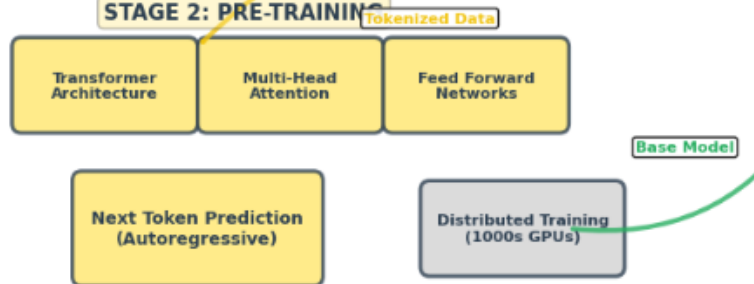
Exercise 5 (Notebook 5): Play with the interactive LLM development pipeline

Modern Large Language Model Development Pipeline

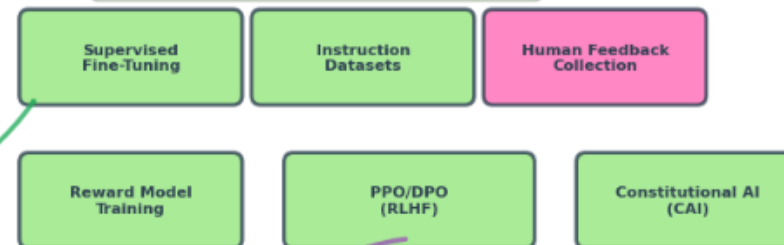
STAGE 1: DATA COLLECTION & PROCESSING



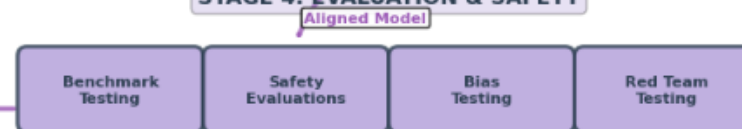
STAGE 2: PRE-TRAINING



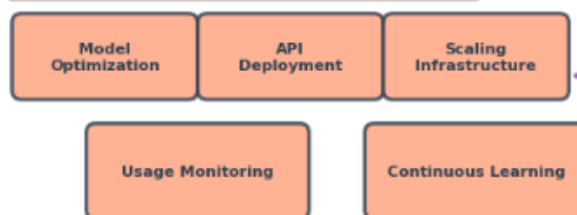
STAGE 3: ALIGNMENT & FINE-TUNING



STAGE 4: EVALUATION & SAFETY



STAGE 5: DEPLOYMENT & MONITORING



Thank you for your Attention!

Please let us know your thoughts:

dartgo.org/feedback

